The sharp product region for Chambolle–Pock four-cycles

Abstract

For every positive extrapolation parameter, we classify the stepsize products admitting an orbit of least period four for the literal primal-first Chambolle–Pock recurrence. Two universal lower bounds are necessary: one is strict and follows from a nonnegative combination of cyclic subgradient inequalities; the other follows from opposite-displacement inequalities and a Gram bound. Three-dimensional indicator polytopes realize every product satisfying both bounds, with an existing saddle point and an explicit initialization. The infimum changes from unattained to attained at a precisely determined parameter, with the crossover itself unattained. The argument retains all temporal Fourier modes and applies to arbitrary finite-dimensional convex problems. A separate zero-parameter corollary identifies a discontinuity of the infimum. This is a finite-period classification, not a convergence theorem. No novelty claim is made.

Keywords: Chambolle–Pock iteration; least-period-four orbits; sharp product region; cyclic convex interpolation; endpoint attainment; polytope realization

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1 The recurrence and convex setting

Let X, Y be finite-dimensional real Hilbert spaces, let $L : X \to Y$ be linear, and let $f : X \to (-\infty, +\infty]$ and $h : Y \to (-\infty, +\infty]$ be proper closed convex functions. For $a > 0$, write

$$\operatorname{prox}_{af}(z) = \arg\min_x \left\{ f(x) + \frac{\|x - z\|^2}{2a} \right\}.$$

Fix $\theta > 0$ and $\tau, \sigma > 0$. We study the paired-state recurrence

$$\begin{aligned} x_{k+1} &= \operatorname{prox}_{\tau f}(x_k - \tau L^* y_k), \\ y_{k+1} &= \operatorname{prox}_{\sigma h}\big(y_k + \sigma L((1+\theta)x_{k+1} - \theta x_k)\big). \end{aligned} \tag{1}$$

The dual update uses the newly computed primal point. There is no outer relaxation or additional stored state. The primal–dual framework originates in Chambolle and Pock [1]; the order and full positive parameter range studied here are specified by (1), rather than attributed to a convergence result in that reference.

2 Complete classification and endpoint behavior

Set

$$a_\theta = \frac{4}{1+2\theta}, \qquad b_\theta = \frac{2}{\sqrt{1+2\theta+2\theta^2}-\theta}, \qquad r = \tau\sigma\|L\|^2. \tag{2}$$

The denominator of b_θ is positive, since $1+2\theta+2\theta^2-\theta^2 = (1+\theta)^2 > 0$.

Theorem 1. *For each $\theta > 0$, the set of products admitting an orbit of least period four for (1), over the entire stated convex class, is exactly*

$$\{r > 0 : r > a_\theta \text{ and } r \geq b_\theta\}. \tag{3}$$

The necessary bounds hold without a saddle-point assumption. Every product in (3) is realized with $X = Y = \mathbb{R}^3$, $L = I$, $\tau = r$, $\sigma = 1$, and indicators of two explicit four-vertex convex hulls, with a saddle pair at $(0,0)$.

The infimum is $\max\{a_\theta, b_\theta\}$. It is unattained for $0 < \theta \leq \sqrt{6}/4$ and attained for $\theta > \sqrt{6}/4$. Thus the exact product set is

$$\begin{cases} (a_\theta, \infty), & 0 < \theta \leq \sqrt{6}/4, \\ [b_\theta, \infty), & \theta > \sqrt{6}/4. \end{cases}$$

These conclusions remain unchanged if existence of a saddle point is required.

The quantifier in the realization statement allows the convex problem to depend on the requested product. The theorem is not a classification of all four-cycles of a fixed problem or of all equality cases.

An orbit of least period four has four distinct paired states and repeats them indefinitely. A four-periodic orbit may instead have least period one or two; these possibilities must be retained in the necessity proof. In particular the strict inequality in (3) cannot be replaced by a weak one, even at the crossover where the two displayed bounds agree.

The rest of the manuscript proves both universal bounds and gives a finite explicit realization for every admissible product. No regular-polygon, planar-orbit, or single-frequency hypothesis is imposed in the lower-bound argument.

3 Actual proximal inequalities and centering

Index a four-periodic orbit by i modulo four. Every component is an output of a proximal map, including phase zero by periodicity; hence all function values at these states are finite. Proximal optimality gives the actual subgradients

$$\begin{aligned} p_i &= (x_{i-1} - x_i)/\tau - L^* y_{i-1} \in \partial f(x_i), \\ z_i &= (y_{i-1} - y_i)/\sigma + L((1+\theta)x_i - \theta x_{i-1}) \in \partial h(y_i). \end{aligned} \tag{4}$$

For every integer displacement j, sum the subgradient inequalities over i. The function values telescope because $i \mapsto i + j$ permutes the phases. Thus

$$\sum_i \langle p_i, x_{i+j} - x_i \rangle \le 0, \qquad \sum_i \langle z_i, y_{i+j} - y_i \rangle \le 0. \tag{5}$$

The values need not be constant along the orbit.

Let $\bar{x} = \frac{1}{4}\sum_i x_i$ and $\bar{y} = \frac{1}{4}\sum_i y_i$, and rescale the centered sequences by

$$U_i = (x_i - \bar{x})/\sqrt{\tau}, \qquad V_i = (y_i - \bar{y})/\sqrt{\sigma}, \qquad K = \sqrt{\tau\sigma}\, L.$$

On fourfold direct sums use $\langle U, W \rangle = \frac{1}{4}\sum_i \langle U_i, W_i \rangle$. The backward shift is $(SU)_i = U_{i-1}$; its adjoint S^* is the forward shift. The same symbol K denotes the diagonal lift. It commutes with shifts, preserves centered sequences, and satisfies $\|K\|^2 = r$.

The constant terms removed by centering are orthogonal to every difference sequence. After rescaling, (5) therefore says precisely

$$\begin{aligned} \langle (S - I)U - K^* S V, ((S^*)^j - I)U \rangle \le 0, \\ \langle (S - I)V + K((1+\theta)I - \theta S)U, ((S^*)^j - I)V \rangle \le 0. \end{aligned} \tag{6}$$

The factor $1/4$ from the direct-sum inner product is positive and does not change the inequalities. These formulas apply in different primal and dual spaces; only the diagonal lift connects them.

4 All temporal Fourier modes and four inequalities

On the centered temporal space, split orthogonally

$$H_1 = \ker(S^2 + I), \qquad H_2 = \ker(S + I), \qquad U = U_1 + U_2, \quad V = V_1 + V_2.$$

For example the projections are $(I - S^2)/2$ and $(I + S^2)/2$ on centered sequences. The first range has $S^2 = -I$. In the second range, $S^2 = I$; centering removes the $S = I$ eigenspace, leaving $S = -I$. Since S^2 is self-adjoint, these projections are orthogonal. Thus the decomposition retains every nonconstant four-point mode, not merely a chosen ansatz.

Put $J = S$ on H_1, so $J^* = -J$, $J^2 = -I$, and J is orthogonal. The lift K preserves both modes and intertwines their shifts. Define

$$\begin{aligned} A &= \|U_1\|^2, & B &= \|V_1\|^2, & E &= \|U_2\|^2, & F &= \|V_2\|^2, \\ c &= \langle V_1, KU_1 \rangle, & q &= \langle V_1, KJU_1 \rangle, & g &= \langle V_2, KU_2 \rangle, & \alpha &= 1 + 2\theta. \end{aligned} \tag{7}$$

Taking $j = 1$ and half of $j = 2$ in (6) gives

$$\begin{aligned} P_1 &= 4E + c - q - 2g \le 0, \\ H_1' &= 4F - c + \alpha q - 2\alpha g \le 0, \\ P_2 &= A - q \le 0, \\ H_2' &= B - (1 + \theta)c + \theta q \le 0. \end{aligned} \tag{8}$$

Here primes distinguish inequality labels from the temporal subspaces.

For clarity, on the first mode $S = J$, $S^* = -J$. The self term for $j = 1$ is zero, while the primal cross term is $-\langle V_1, KS^*(S^* - I)U_1 \rangle = c - q$. The dual cross polynomial is $(S - I)((1 + \theta)I - \theta S) = -I + \alpha J$, giving $-c + \alpha q$. On the alternating mode both differences are $-2I$, giving $4E - 2g$ and $4F - 2\alpha g$. For $j = 2$ the alternating differences vanish; the first-mode differences are $-2I$. Dividing these two inequalities by two yields $A - q$ and $B - (1 + \theta)c + \theta q$, as claimed.

5 The strict universal bound

A positive weighted combination of the four necessary inequalities has an exact nonnegative certificate. From (8),

$$\begin{aligned} 0 \ &\geq \alpha P_1 + H_1' + \frac{2\theta}{1+\theta} H_2' + \frac{2\theta^2}{1+\theta} P_2 \\ &= 4F + 4\alpha E - 4\alpha g + \frac{2\theta}{1+\theta} B + \frac{2\theta^2}{1+\theta} A \\ &= 4 \left\| V_2 - \frac{\alpha}{2} K U_2 \right\|^2 + \alpha(4 - \alpha r) E \\ &\qquad + \alpha^2 \big(rE - \|KU_2\|^2 \big) + \frac{2\theta}{1+\theta} B + \frac{2\theta^2}{1+\theta} A. \end{aligned} \tag{9}$$

To verify the last identity, expand the square as $4F - 4\alpha g + \alpha^2 \|KU_2\|^2$ and cancel the norm terms. No estimate is hidden in the identity.

Suppose $r \leq 4/\alpha$. Every term on the last two lines of (9) is nonnegative: $E, A, B \geq 0$, $\|KU_2\|^2 \leq rE$, and $\theta > 0$ makes both final coefficients strictly positive. Therefore $A = B = 0$. The centered orbit has only the alternating mode, so $S^2 U = U$ and $S^2 V = V$. Restoring the means and scales shows that the original paired orbit has period dividing two. It cannot have least period four. Hence every least-period-four orbit satisfies

$$r > \frac{4}{\alpha} = a_\theta. \tag{10}$$

If $r < 4/\alpha$, the same certificate also forces $E = 0$ and then $V_2 = (\alpha/2) K U_2 = 0$. Thus every four-periodic orbit in that strict range is constant. At equality only period dividing two is forced; constancy is not asserted there.

This proof does not divide by $\|L\|$, by a mode norm, or by an orbit amplitude. It includes $L = 0$ and both possible vanishing Fourier modes. It uses neither saddle existence nor strict convexity. The positivity assumption $\theta > 0$ is essential to the conclusion $A = B = 0$; the statement here does not include $\theta = 0$.

6 The second bound from opposite displacements

The last two inequalities of (8) give

$$q \geq A, \qquad B \leq (1+\theta)c - \theta q. \tag{11}$$

For an orbit of least period four, $A > 0$. Indeed if $A = 0$, then $U_1 = 0$, so $c = q = 0$ and (11) forces $B = 0$. Only the alternating mode and the means remain, contradicting least period four.

Because $KJ = JK$, the vectors KU_1 and KJU_1 are orthogonal and have the same norm: J is skew-adjoint and orthogonal. If their norm is zero, $c = q = 0$, contradicting $q \geq A > 0$. Otherwise divide both by their common positive norm and apply Bessel's inequality to V_1. This proves

$$c^2 + q^2 \leq \|KU_1\|^2 B \leq rAB. \tag{12}$$

The displayed inequality itself is also valid in the zero-norm case, with both left terms zero. Thus no degeneracy is omitted.

Now $A > 0$ permits $u = c/A$ and $v = q/A \geq 1$. By (11)–(12),

$$u^2 + v^2 \leq r((1+\theta)u - \theta v).$$

Complete the square to obtain

$$\left(u - \frac{r(1+\theta)}{2}\right)^2 + v^2 + r\theta v \leq \frac{r^2(1+\theta)^2}{4}.$$

Since $v \geq 1$ and $r, \theta > 0$, its left side is at least $1 + r\theta$. Consequently

$$(1+\theta)^2 r^2 - 4\theta r - 4 \geq 0.$$

This quadratic has one negative and one positive root. With $r > 0$ the necessary condition is

$$r \geq \frac{2(\theta + \sqrt{\theta^2 + (1+\theta)^2})}{(1+\theta)^2} = b_\theta. \tag{13}$$

The rationalization is valid because $(1+\theta)^2 > 0$. The first bound already gives $r > 0$; alternatively $q \geq A > 0$ and (12) exclude $r = 0$. Both bounds are therefore universal and have accounted for every division and zero-coupling case.

7 An explicit finite construction for every product

Fix $\theta > 0$ and any $r > a_\theta$ with $r \geq b_\theta$. All notation in this construction is defined afresh; the scalars below specify actual Euclidean vertices. Set

$$\begin{aligned} \alpha &= 1 + 2\theta, & c &= \frac{1+\theta}{2}, \quad q = \frac{1}{r}, \\ d &= c - q, & B &= c^2 + q^2, \quad \delta = \frac{4(\alpha r - 4)}{r^2}, \\ C_1 &= \frac{8d}{r} - (\alpha+1)c + 2\alpha q, \\ C_3 &= \frac{8d}{r} + 2B - 2\alpha c + (\alpha - 1)q. \end{aligned} \tag{14}$$

Here $\delta > 0$ by the strict bound. Moreover

$$(1 + 2\theta)^2 - (1 + 2\theta + 2\theta^2) = 2\theta(1 + \theta) > 0.$$

Both compared square roots are positive, so $\sqrt{1 + 2\theta + 2\theta^2} \leq 1 + 2\theta$. Hence $b_\theta \geq 2/(1+\theta)$ and $r \geq 2/(1 + \theta)$, proving $d \geq 0$.

Choose the finite integer

$$m = \left\lceil \sqrt{1 + \frac{|C_1| + |C_3| + d^2}{\delta}} \right\rceil, \qquad g = \frac{2m^2}{r} + \frac{d}{2}. \tag{15}$$

In particular $m \geq 1$ and

$$m^2 \geq 1 + \frac{|C_1| + |C_3| + d^2}{\delta}.$$

Define the orthogonal map and vertices

$$\begin{aligned} T(u, v, w) &= (-v, u, -w), \\ \xi_i &= T^i(1, 0, m), \quad \eta_i = T^i(c, -q, g/m), \qquad 0 \leq i < 4. \end{aligned} \tag{16}$$

Let $C = \operatorname{conv}\{\xi_0, \xi_1, \xi_2, \xi_3\}$ and $D = \operatorname{conv}\{\eta_0, \eta_1, \eta_2, \eta_3\}$. Use

$$X = Y = \mathbb{R}^3, \quad L = I, \quad \tau = r, \quad \sigma = 1, \quad f = \iota_C, \quad h = \iota_D.$$

Here ι_Q is zero on Q and $+\infty$ elsewhere. The two nonempty finite convex hulls are compact, and their indicators are proper closed convex. Their proximal maps are the Euclidean projections, independently of the positive parameter. All expressions are finite; the only amplitude division is by $m \geq 1$.

8 All twenty four normal cone tests

For a closed convex set Q, a point $z \in Q$ equals $P_Q(w)$ if and only if $\langle w - z, z' - z \rangle \le 0$ for all $z' \in Q$. Sufficiency follows by expanding squared distances; necessity follows by testing the segment from z to z' and letting its parameter decrease to zero. Projection is unique. For a finite convex hull it suffices to test the vertices, since the expression is affine in z'.

For the transition from phase $i-1$ to phase i, put

$$\begin{aligned} p_i &= (\xi_{i-1} - \xi_i)/r - \eta_{i-1}, \\ z_i &= \eta_{i-1} - \eta_i + (1+\theta)\xi_i - \theta\xi_{i-1}. \end{aligned} \tag{17}$$

The required conditions are $p_i \in N_C(\xi_i)$ and $z_i \in N_D(\eta_i)$. Write $E = m^2$, $F = g^2/E$. Direct multiplication at phase zero gives, in target order $1, 2, 3$,

$$\begin{aligned} \langle p_0, \xi_1 - \xi_0 \rangle &= 4E/r + c - q - 2g = 0, \\ \langle p_0, \xi_2 - \xi_0 \rangle &= 2/r - 2q = 0, \\ \langle p_0, \xi_3 - \xi_0 \rangle &= 2/r + 4E/r - c - q - 2g = -2d \le 0. \end{aligned} \tag{18}$$

The dual products are

$$\begin{aligned} \langle z_0, \eta_1 - \eta_0 \rangle &= 4F - c + \alpha q - 2\alpha g = -\delta E + C_1 + d^2/E, \\ \langle z_0, \eta_2 - \eta_0 \rangle &= 2[B - (1+\theta)c + \theta q] = 2[q^2 + \theta q - (1+\theta)^2/4], \\ \langle z_0, \eta_3 - \eta_0 \rangle &= 2B + 4F - \alpha c - q - 2\alpha g = -\delta E + C_3 + d^2/E. \end{aligned} \tag{19}$$

The middle expression is nonpositive exactly when $r \ge b_\theta$: substitute $q = 1/r$ and multiply by $r^2 > 0$ to recover the quadratic in (13). For $j = 1, 3$, (15) and $E \ge 1$ give

$$-\delta E + C_j + d^2/E \le -\delta - (|C_1| + |C_3| + d^2) + C_j + d^2 \le -\delta < 0.$$

Finally T takes each vertex and each normal to the next, preserves both hulls and all inner products, and satisfies $T^4 = I$. Thus these six tests prove all $4 \cdot 3 \cdot 2 = 24$ nontrivial vertex inequalities. Self-target tests are zero. They are actual normal-cone certificates for the specified convex functions.

9 Initialization saddle point and exact product

The primal residual in the projection update is rp_i, a positive multiple of the verified normal; the dual residual is z_i. Consequently the literal recurrence initialized at

$$(x_0, y_0) = (\xi_0, \eta_0)$$

satisfies $(x_k, y_k) = (\xi_{k \bmod 4}, \eta_{k \bmod 4})$ for every $k \geq 0$, by induction using (17). No extrapolated-state initialization or hidden prehistory is needed. The four primal planar coordinates $(1,0), (0,1), (-1,0), (0,-1)$ are distinct, so the least period is exactly four.

Both sums $\sum_i \xi_i$ and $\sum_i \eta_i$ vanish: the planar quarter-turn terms and the alternating third coordinates each sum to zero. Thus $0 \in C \cap D$ and

$$-L^*0 = 0 \in N_C(0), \qquad L0 = 0 \in N_D(0).$$

These are the saddle conditions for $\mathcal{F}(x,y) = f(x) + \langle Lx, y\rangle - h(y)$. Equivalently, on the effective domains $C \times D$, $\mathcal{F}(0,y) = 0 = \mathcal{F}(x,0)$, and the extended-value saddle inequalities hold when either argument leaves its effective domain. If one writes $h = g^*$, take $g = h^*$; closed convexity gives $g^* = h$.

The norm of $L = I$ in the stated Euclidean spaces is exactly one. Hence $\tau\sigma\|L\|^2 = r$, with the prescribed product rather than a surrogate Gram parameter. When θ, r are rational, c, q, d, g and all vertex coordinates are rational because m is an integer. Any Gram matrix formed from these vertices is positive semidefinite by its explicit $\mathbb{R}^3$ factorization, not by a numerical eigenvalue test.

This completes the sufficiency proof for every product in (3), including $r = b_\theta$ whenever the strict first bound permits it. The amplitude bound may be conservative, but is always finite for an admissible product. A limit of polytopes or an infinite-dimensional construction is unnecessary.

For a simple rational example, $\theta = 1$, $r = 17/10$ allows $m = 2$. Then

$$\xi_0 = (1, 0, 2), \qquad \eta_0 = (1, -10/17, 167/68),$$

with the other phases obtained by T. The actual normal products are verified exactly by the accompanying script.

10 The separate zero extrapolation endpoint

Corollary 10.1. *At $\theta = 0$, the exact set of products admitting least period four is $[2, \infty)$, also when a saddle point is required. The minimum 2 is attained. In contrast, the infimum for positive θ tends to 4 as $\theta \downarrow 0$.*

Proof. The cyclic inequalities, centering, Fourier decomposition, and opposite-displacement bounds (11) remain valid at zero. They give $q \geq A$ and $c \geq B$. Least period four requires $A, B > 0$. Indeed, $A = 0$ makes $c = q = 0$ and forces $B = 0$. Likewise $B = 0$ makes $c = q = 0$ and forces $A = 0$. Either case leaves period dividing two. The Bessel inequality (12) therefore yields

$$rAB \geq c^2 + q^2 \geq B^2 + A^2 \geq 2AB.$$

Division by $AB > 0$ proves $r \geq 2$. This also excludes zero coupling; no division by $\|L\|$ is used. The final inequality is $A^2 + B^2 - 2AB = (A - B)^2 \geq 0$.

For every $r \geq 2$, work in $\mathbb{R}^2$ with the orthogonal quarter-turn $R(u, v) = (-v, u)$ and define

$$\xi_i = R^i(1, 0), \qquad \eta_i = R^i(1/2, -1/2), \qquad 0 \leq i < 4.$$

Take $C = \operatorname{conv}\{\xi_i\}$, $D = \operatorname{conv}\{\eta_i\}$, $f = \iota_C$, $h = \iota_D$, $L = I$, $\tau = r$, $\sigma = 1$. Put $q_0 = 1/r \in (0, 1/2]$. The required phase-zero normals are

$$p_0 = (\xi_3 - \xi_0)/r - \eta_3 = (1/2 - q_0, 1/2 - q_0), \qquad z_0 = \eta_3 - \eta_0 + \xi_0 = 0.$$

The three primal vertex products, in order $1, 2, 3$, are

$$\big(\langle p_0, \xi_j - \xi_0 \rangle\big)_{j=1}^{3} = (0, 2q_0 - 1, 2q_0 - 1) \leq 0.$$

All three dual products are zero. Rotation gives all 24 vertex normal tests, so the projection criterion proves every actual transition. Initialization at (ξ_0, η_0) gives the literal orbit, of least period four because its primal vertices are distinct. Both hulls have zero vertex mean, yielding the saddle pair $(0, 0)$. Their indicators are proper closed convex and $\|I\| = 1$, so the product is exactly r. At $r = 2$ all primal normals also vanish; the same finite construction attains the endpoint.

Finally $a_\theta \to 4$ and $b_\theta \to 2$ as $\theta \downarrow 0$, so Theorem 1 gives a right-hand infimum limit of 4. The strict certificate used positive coefficients proportional to θ and θ^2, which vanish at zero; substituting zero into its strict conclusion is invalid. ∎

11 Infimum attainment and scope

The necessity and construction together establish (3). Comparing the two boundaries gives

$$b_\theta > a_\theta \iff \sqrt{1 + 2\theta + 2\theta^2} < 2\theta + \tfrac{1}{2}$$
$$\iff \theta^2 > \tfrac{3}{8} \iff \theta > \frac{\sqrt{6}}{4}.$$

Both sides of the comparison before squaring are positive, so the equivalences preserve signs. At equality $a_\theta = b_\theta$, but (10) still excludes the endpoint. Below the crossover the admissible set is (a_θ, ∞); above it the construction applies even at $b_\theta > a_\theta$, giving $[b_\theta, \infty)$.

In particular, at $\theta = 1/8$ the unattained infimum is $16/5$. At $\theta = 1/4$ it is $8/3$, also unattained. At $\theta = 1$ the minimum is the golden ratio,

$$b_1 = \frac{1 + \sqrt{5}}{2}.$$

The script also verifies an explicit realization of this attained endpoint with $m = 2$. These examples illustrate both kinds of boundary without replacing the all-parameter proof.

The theorem concerns only least-period-four orbits of (1). It does not exclude other periods or nonperiodic failure of convergence, identify a universal convergence threshold, assert attraction or robustness, or treat a different update order. The realized cycles are bounded and their ordinary Cesàro averages converge to $(0, 0)$, since each block of four states sums to zero. Full iterates nevertheless do not converge.

11.1 Reproducibility

Run `python3 verify_exact.py` with Python 3 and SymPy. The portable verifier checks the four Fourier coefficient identities, the nonnegative certificate, the completed square and exact root, all 24 normal products, centroids, distinct phases, rational Gram factors, an attained algebraic endpoint, and all normal tests for the separate zero-parameter corollary. The proof of signs over the full parameter region is given above; the exact examples in the script are supplements rather than numerical evidence for that region.

Run `bash build.sh` with pdfLaTeX to reproduce the manuscript. The source package contains this self-contained proof, its unchanged reference-style file, the build script, and the verifier.

References

[1] A. Chambolle and T. Pock, A first-order primal-dual algorithm for convex problems with applications to imaging, *Journal of Mathematical Imaging and Vision* **40** (2011), 120–145. doi:10.1007/s10851-010-0251-1.